Lab Experiment-14

Aim

To write a Python program to check grammatical agreement in sentences using Context-Free Grammar (CFG) rules and verify whether the subject and verb agree in number (singular or plural).

Algorithm

1. Start the program.
2. Import the CFG and ChartParser modules from the NLTK library.
3. Define CFG production rules for singular and plural nouns and verbs.
4. Define grammar rules for sentence formation using subject-verb agreement.
5. Create a parser using the defined grammar.
6. Get a sentence as input from the user.
7. Convert the sentence into lowercase words.
8. Check whether the sentence can be parsed using the CFG.
9. If a valid parse tree is generated, display "Sentence is grammatically correct."
10. Otherwise, display "Sentence is grammatically incorrect."
11. Stop the program.

Python Code

```
import nltk

from nltk import CFG

from nltk.parse import ChartParser

grammar = CFG.fromstring("""

  S -> NP_SG VP_SG

  S -> NP_PL VP_PL
```

```
    NP_SG -> Det_SG N_SG

    NP_PL -> Det_PL N_PL

    VP_SG -> V_SG

    VP_PL -> V_PL

    Det_SG -> 'the' | 'a'

    Det_PL -> 'the'

    N_SG -> 'boy' | 'girl' | 'dog'

    N_PL -> 'boys' | 'girls' | 'dogs'

    V_SG -> 'runs' | 'eats' | 'plays'

    V_PL -> 'run' | 'eat' | 'play'

""")

parser = ChartParser(grammar)

sentence = input("Enter a sentence: ").lower()

words = sentence.split()

trees = list(parser.parse(words))

if trees:

    print("\nSentence:", sentence)

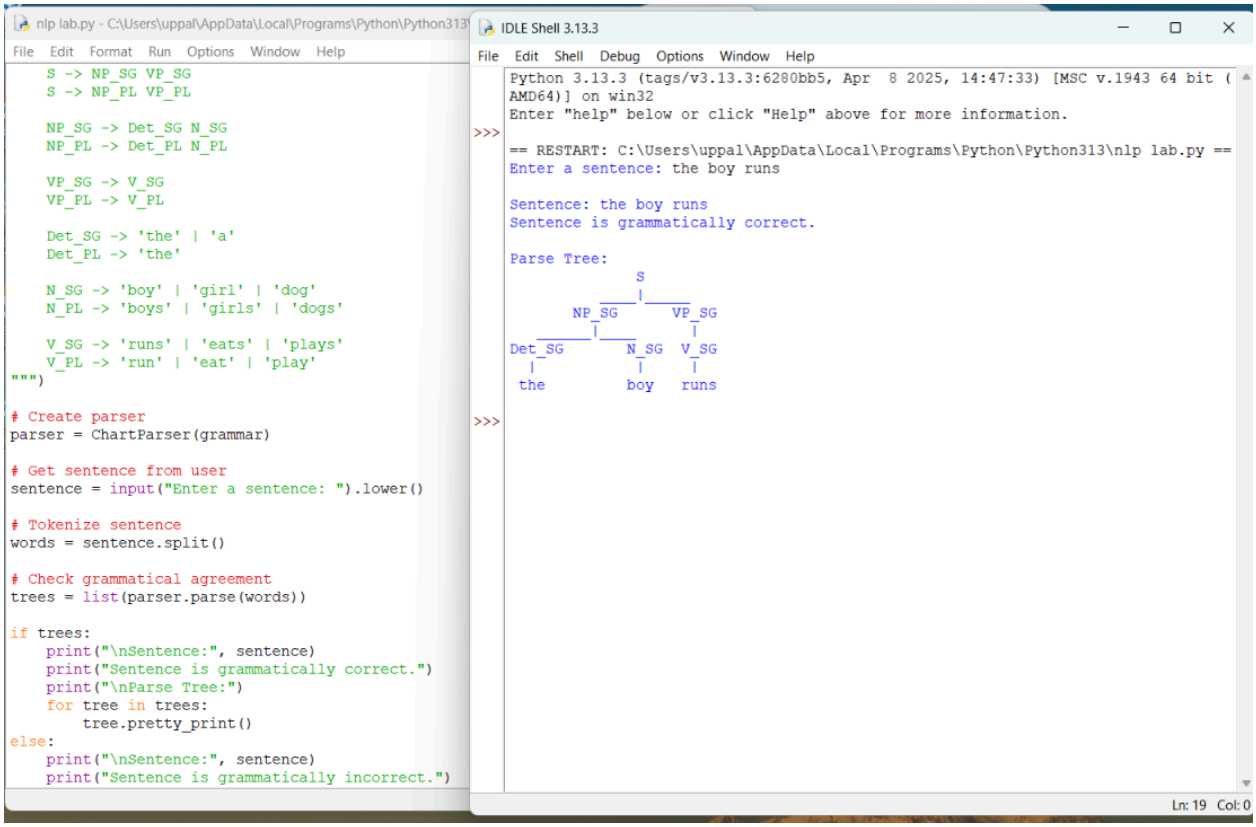
    print("Sentence is grammatically correct.")

    print("\nParse Tree:")

    for tree in trees:
```

```
    tree.pretty_print()
else:
    print("\nSentence:", sentence)
    print("Sentence is grammatically incorrect.")
```

Output:



Result

Thus, the Python program was successfully implemented to check subject-verb agreement in sentences using Context-Free Grammar (CFG) rules. It correctly identifies grammatically valid and invalid sentences based on singular and plural agreement.